

3.5.1 Current electricity

3.5.1.1 Basics of electricity

Content	Opportunities for skills development
Electric current as the rate of flow of charge; potential difference as work done per unit charge. $I = \frac{\Delta Q}{\Delta t}, \quad V = \frac{W}{Q}$ Resistance defined as $R = \frac{V}{I}$	AT b, f Students can construct circuits from the range of components.

3.5.1.2 Current–voltage characteristics

Content	Opportunities for skills development
For an ohmic conductor, semiconductor diode, and filament lamp. Ohm's law as a special case where $I \propto V$ under constant physical conditions. Unless specifically stated in questions, ammeters and voltmeters should be treated as ideal (having zero and infinite resistance respectively). Questions can be set where either I or V is on the horizontal axis of the characteristic graph.	

3.5.1.3 Resistivity

Content	Opportunities for skills development
Resistivity, $\rho = \frac{RA}{L}$ Description of the qualitative effect of temperature on the resistance of metal conductors and thermistors. Only negative temperature coefficient (ntc) thermistors will be considered. Applications of thermistors to include temperature sensors and resistance–temperature graphs. Superconductivity as a property of certain materials which have zero resistivity at and below a critical temperature which depends on the material. Applications of superconductors to include the production of strong magnetic fields and the reduction of energy loss in transmission of electric power. Critical field will not be assessed.	MS 3.2, 4.3 / PS 1.2 / AT a, b, f, g Investigation of the variation of resistance of a thermistor with temperature.
Required practical 5: Determination of resistivity of a wire using a micrometer, ammeter and voltmeter.	

3.5.1.4 Circuits

Content	Opportunities for skills development
Resistors: <i>in series</i>, $R_T = R_1 + R_2 + R_3 + \dots$ <i>in parallel</i>, $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$ <i>Energy and power equations:</i> $E = IVt$; $P = IV = I^2R = \frac{V^2}{R}$ The relationships between currents, voltages and resistances in series and parallel circuits, including cells in series and identical cells in parallel. Conservation of charge and conservation of energy in dc circuits.	MS 0.3 / PS 4.1 / AT a, b, f, g Students can construct circuits with various component configurations and measure currents and potential differences.

3.5.1.5 Potential divider

Content	Opportunities for skills development
The potential divider used to supply constant or variable potential difference from a power supply. The use of the potentiometer as a measuring instrument is not required. Examples should include the use of variable resistors, thermistors, and light dependent resistors (LDR) in the potential divider.	MS 3.2 / PS 4.1 / AT f Students can investigate the behaviour of a potential divider circuit. MS 3.2 / AT g Students should design and construct potential divider circuits to achieve various outcomes.

3.5.1.6 Electromotive force and internal resistance

Content	Opportunities for skills development
$\varepsilon = \frac{E}{Q}, \varepsilon = I(R + r)$ Terminal pd; emf Students will be expected to understand and perform calculations for circuits in which the internal resistance of the supply is not negligible.	
Required practical 6: Investigation of the emf and internal resistance of electric cells and batteries by measuring the variation of the terminal pd of the cell with current in it.	MS 3.1, 3.3 / PS 2.2, 3.1 / AT f